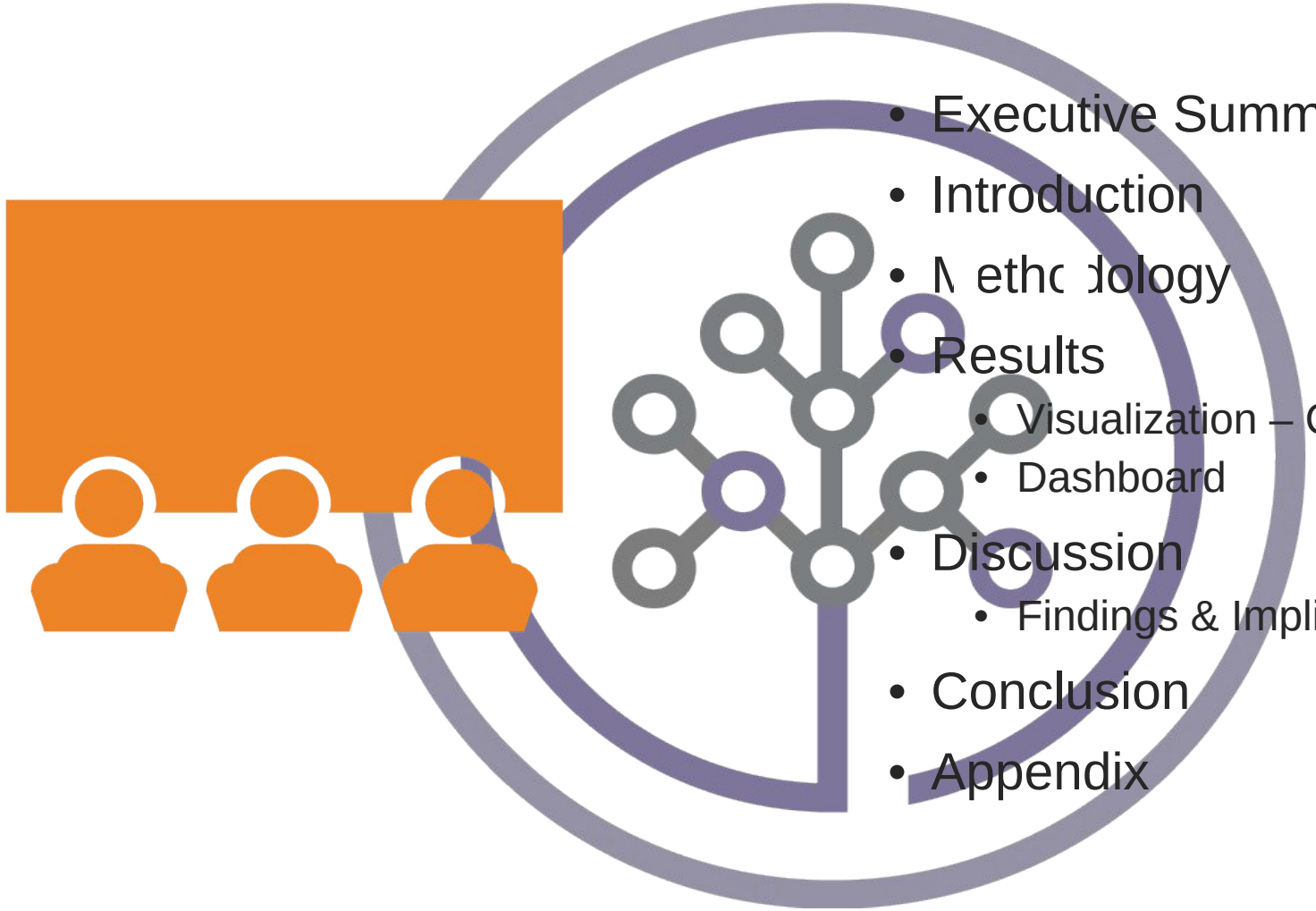
Programming Languages and Technologies Survey Report.

Learner's Name: Moath Alshayiqi

Date: March 2026.

© IBM Corporation. All rights reserved.

OUTLINE



- Executive Summary
- Introduction
- Methodology
- Results
  - Visualization – Charts
  - Dashboard
- Discussion
  - Findings & Implications
- Conclusion
- Appendix

EXECUTIVE SUMMARY

- JavaScript and Python are the most utilized programming languages among the surveyed developers.
- PostgreSQL and MySQL dominate the database market, showing a strong preference for relational systems.
  - Sub Point 1
  - Sub Point 2
  - Sub Point 3
- AWS remains the top cloud platform currently in use by a significant margin.

INTRODUCTION

- Report Purpose

To analyze and visualize the results of the 2023 Stack Overflow Developer Survey to identify current industry standards.

- Target Audience

Technology Leaders and CTOs looking to align their tech stacks with global trends.

- Value Proposition

Provides data-driven insights to reduce risk in technology adoption and strategic planning.

Ensures high accuracy in reporting by applying rigorous data filtering and visualization techniques

METHODOLOGY

- Data Sources & Collection

Data assets were integrated into Google Looker Studio for interactive visualization and analysis.

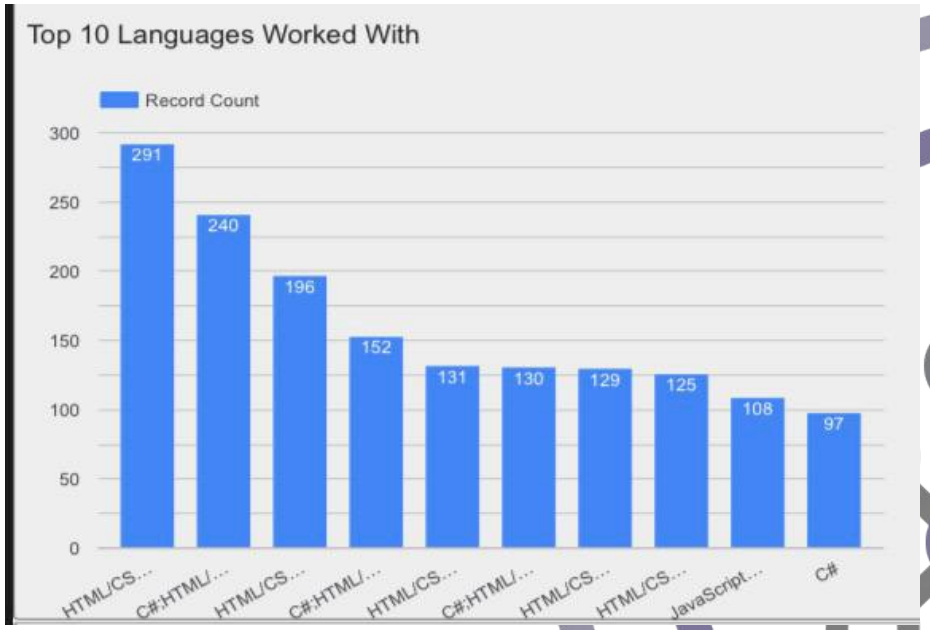
- Analysis Tools & Visualization

Employed advanced visualization types including Tree Maps, Word Clouds, and Geo Charts to represent complex hierarchical and geographic data.

- Data Wrangling & Cleaning Steps

Applied strict filters to exclude null values, ensuring that visualizations represent accurate "Top 10" trends without data gaps.

RESULTS



Current Usage Overview

The bar chart highlights the **Top 10 Programming Languages** that developers have worked with during the current year.

JavaScript and **Python** emerge as the clear leaders, reflecting their versatility in web development and data science.

HTML/CSS and **SQL** maintain high rankings, showing the continued importance of foundational web and database skills.

Key Findings & Analysis

There is a visible dominance of **Web-centric languages**, which aligns with the global shift toward cloud-based applications.

Python's high position suggests a significant portion of the community is engaged in AI, machine learning, or automated scripting.

The data was filtered to **exclude null values**, ensuring the rankings represent verified responses from active developers.

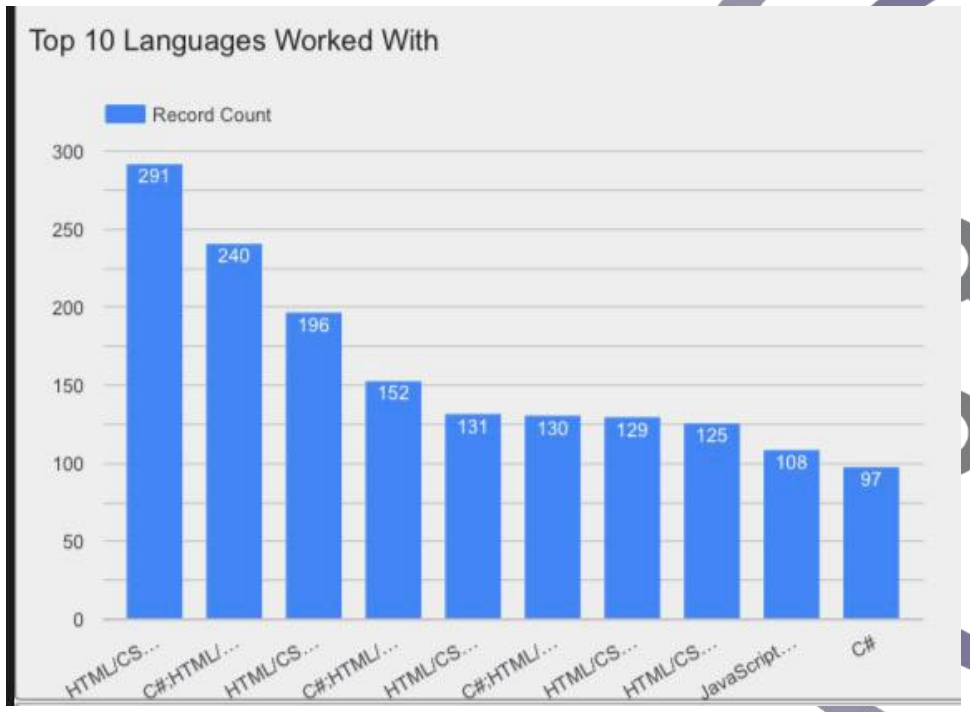
Professional Implications

For job seekers, proficiency in **JavaScript and Python** remains a critical requirement for meeting current market demands.

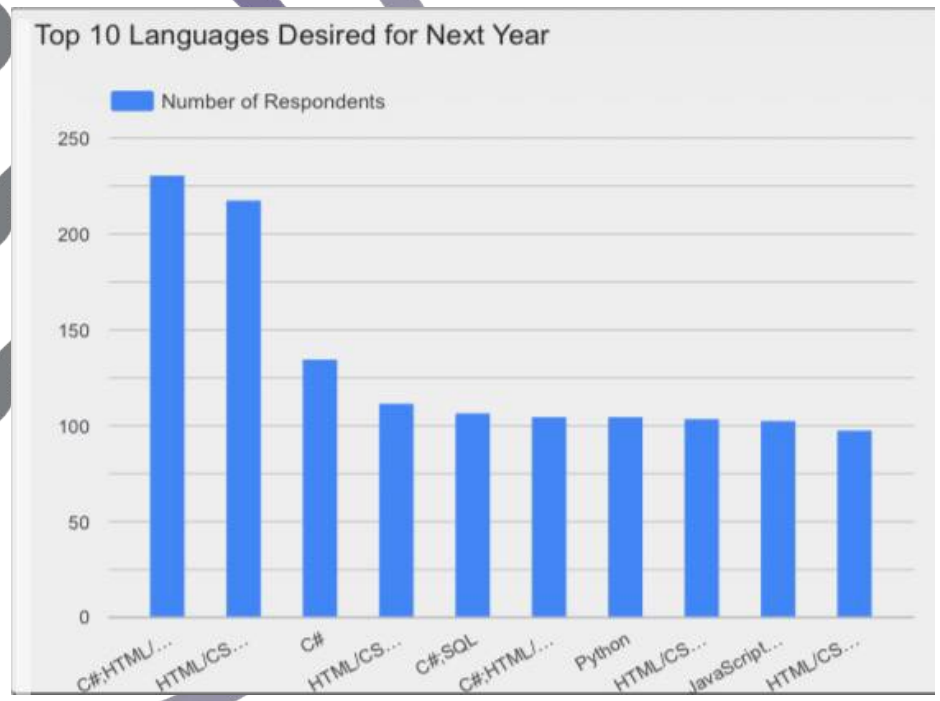
Organizations should continue investing in these ecosystems as they possess the largest talent pools and community support.

PROGRAMMING LANGUAGE TRENDS

Current Year



Next Year



PROGRAMMING LANGUAGE TRENDS - FINDINGS & IMPLICATIONS

Findings

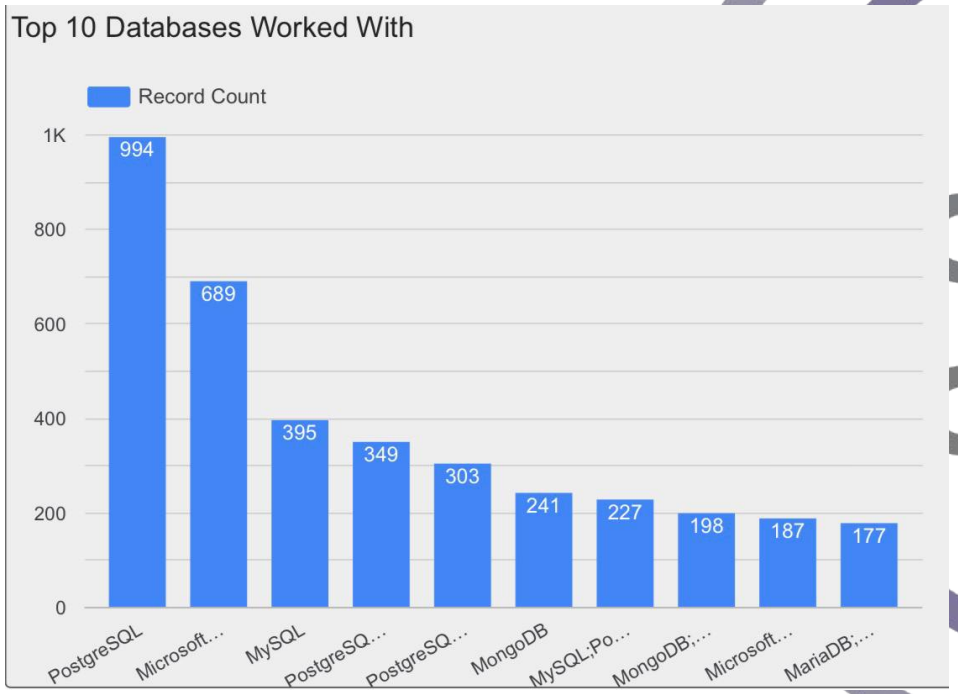
- JavaScript and Python maintain clear dominance as the most utilized languages currently and are the top desired for the upcoming year
- HTML/CSS and SQL follow closely in rankings, emphasizing the critical importance of core web and database management skills in the industry.
- There is a highly stable transition trend, as developers primarily aim to master or continue using the same high-demand languages they currently work with

Implications

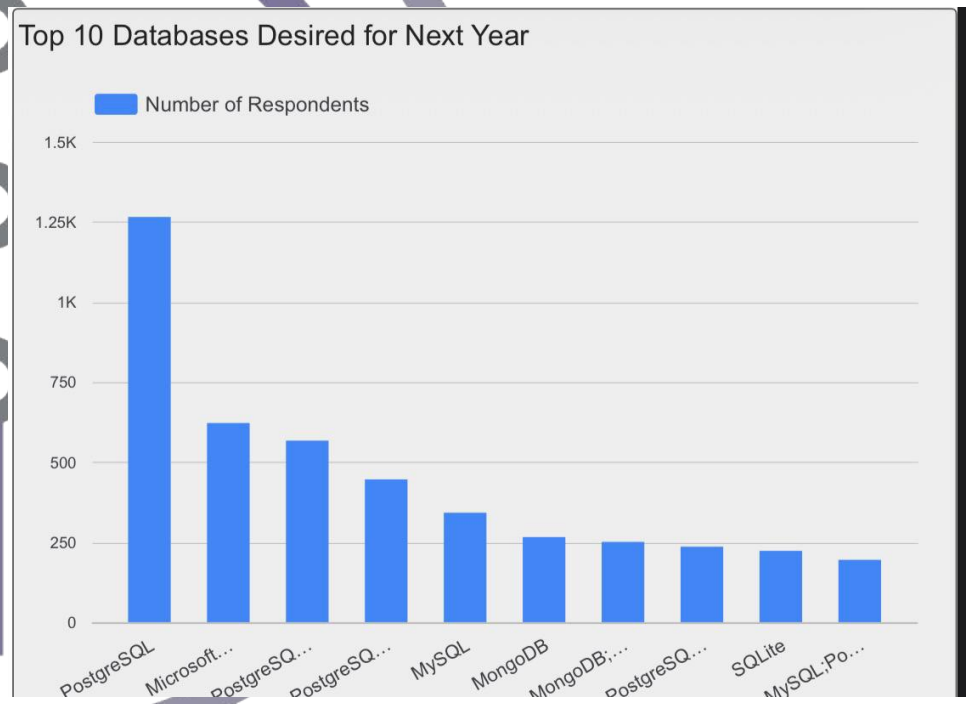
- Market Competitiveness: For job seekers and professionals, maintaining advanced proficiency in Python and JavaScript is essential to meet the highest market demands
- Strategic Upskilling: Educational and corporate training programs should prioritize these foundational languages to ensure the workforce remains aligned with global technology standards.
- Cloud & Web Synergy: The strong interest in HTML and SQL alongside Python suggests a growing industry need for Full-Stack and Data-centric capabilities.

DATABASE TRENDS

Current Year



Next Year



DATABASE TRENDS - FINDINGS & IMPLICATIONS

Findings

- PostgreSQL and MySQL are the clear leaders in current professional usage, reflecting a strong industry preference for reliable, open-source relational systems
- High Future Demand: PostgreSQL continues to be the most desired database for the next year, showing that developers value its advanced features and scalability
- Stable Ecosystem: There is a high correlation between current usage and future desire, suggesting that the current database ecosystem is mature and stable

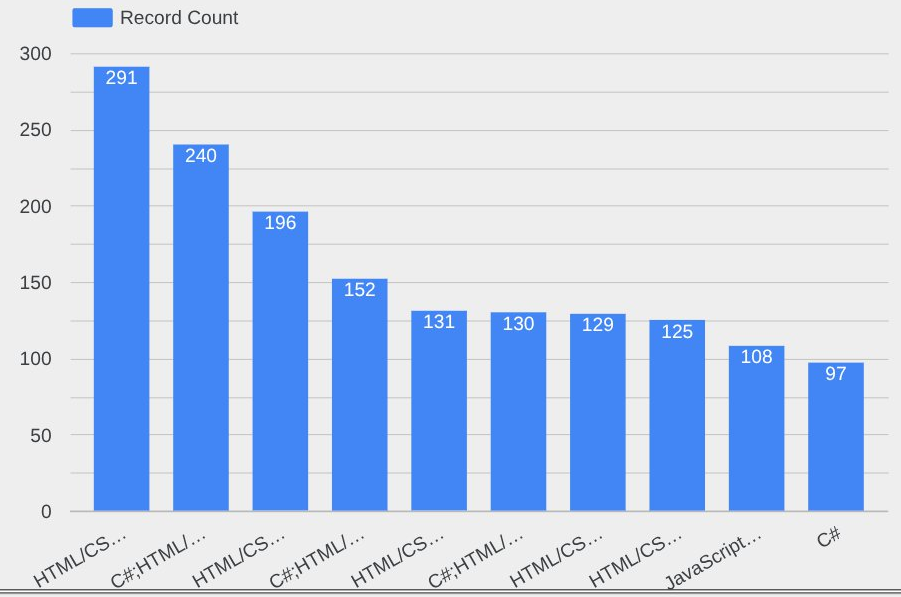
Implications

- Standardization: Organizations should consider standardizing on PostgreSQL or MySQL to benefit from the large talent pool and extensive community support
- Strategic Planning: For data architects, the continued dominance of these systems confirms they are safe, long-term investments for enterprise applications
- Skill Development: Learning SQL and relational database management remains a top priority for developers to stay aligned with both current and future market needs.

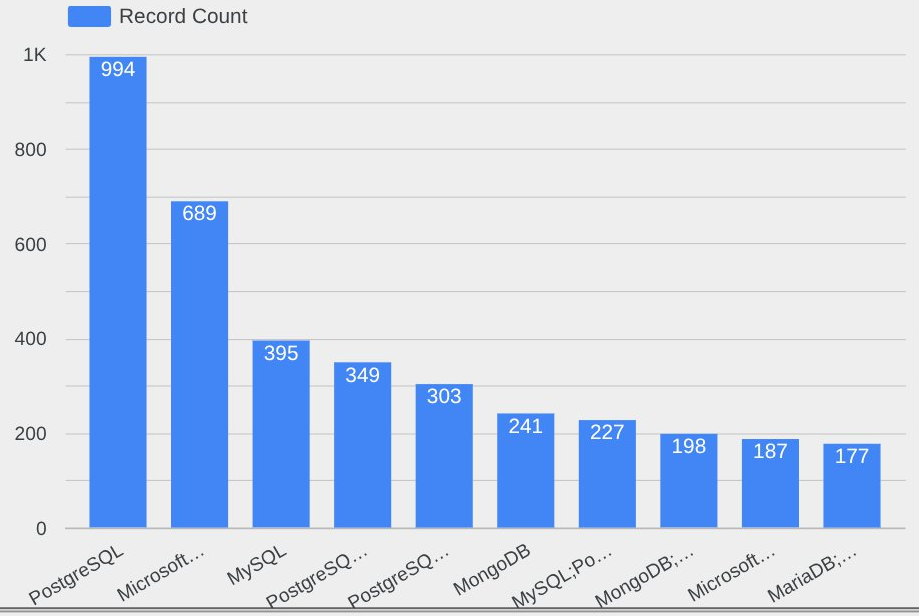
DASHBOARD

https://lookerstudio.google.com/reporting/98ab1ff5-f2d5-489a-8c4e-0c7251b5c90d

Top 10 Languages Worked With

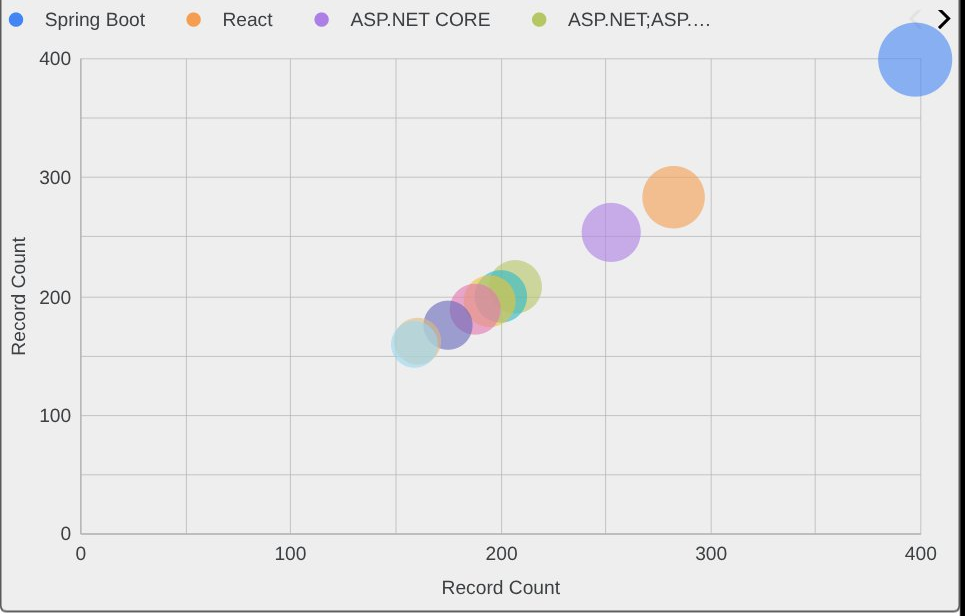


Top 10 Databases Worked With

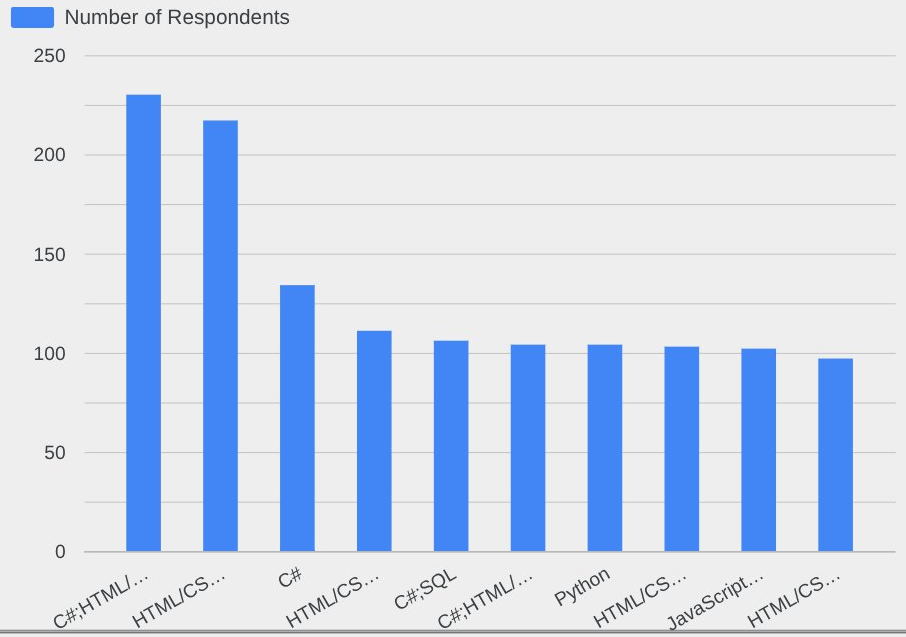


Top 10 Web Frameworks.

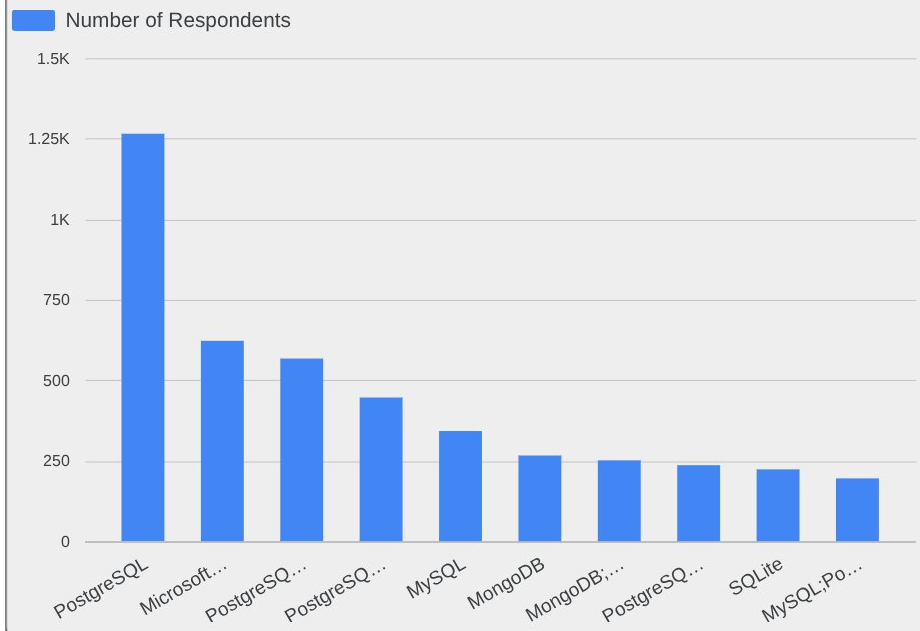
Top 10 Web Frameworks Worked With



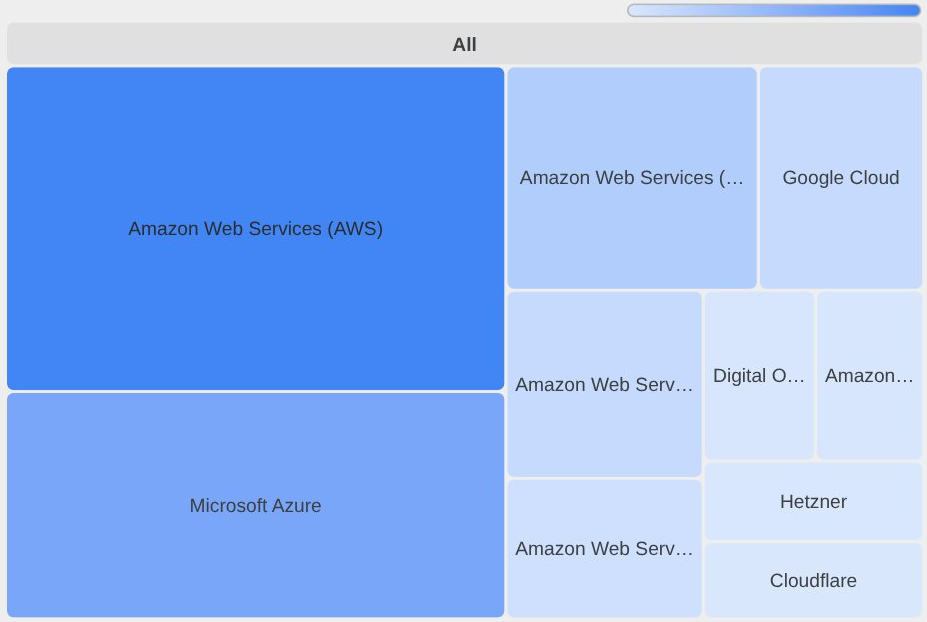
Top 10 Languages Desired for Next Year



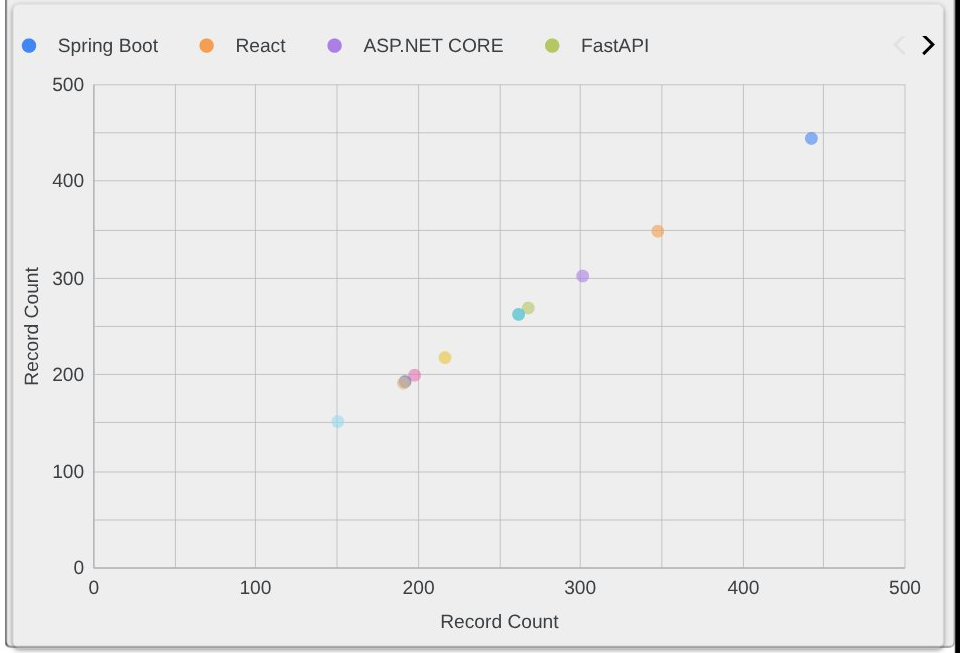
Top 10 Databases Desired for Next Year



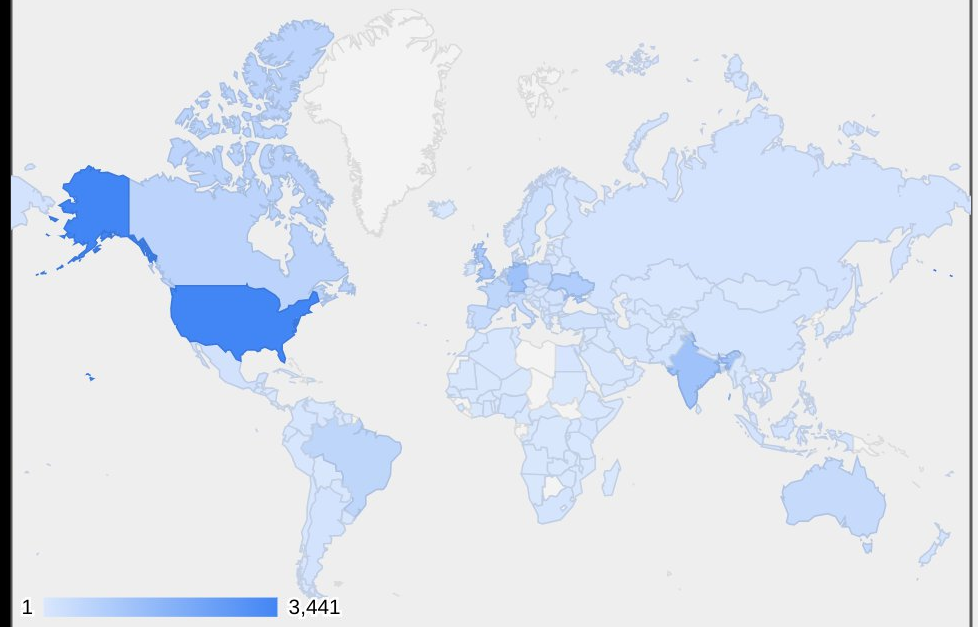
Top 10 Platforms Desired for Next Year



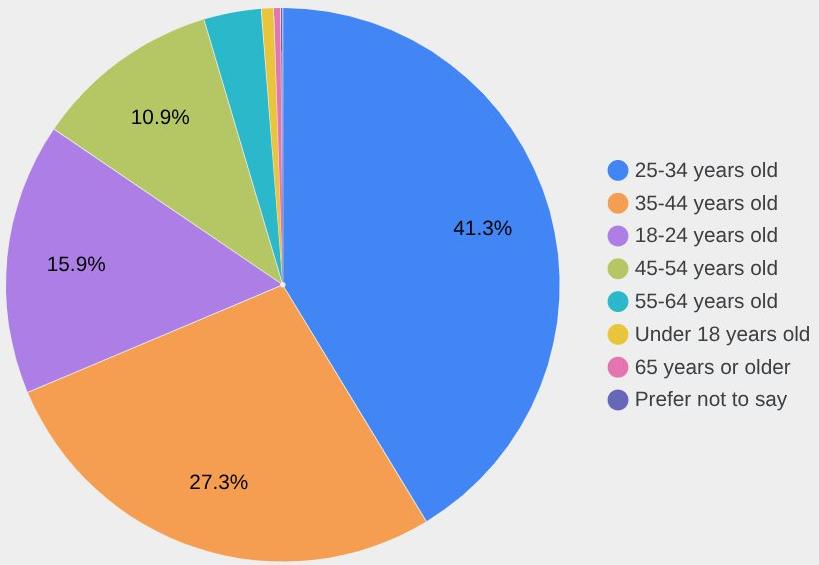
Top 10 Web Frameworks Desired for Next Year



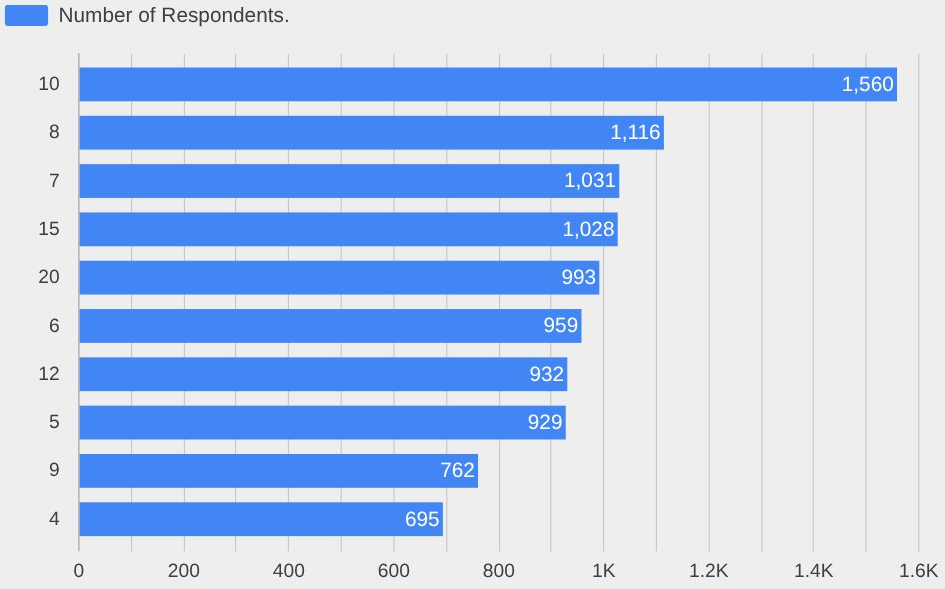
Respondents by Country



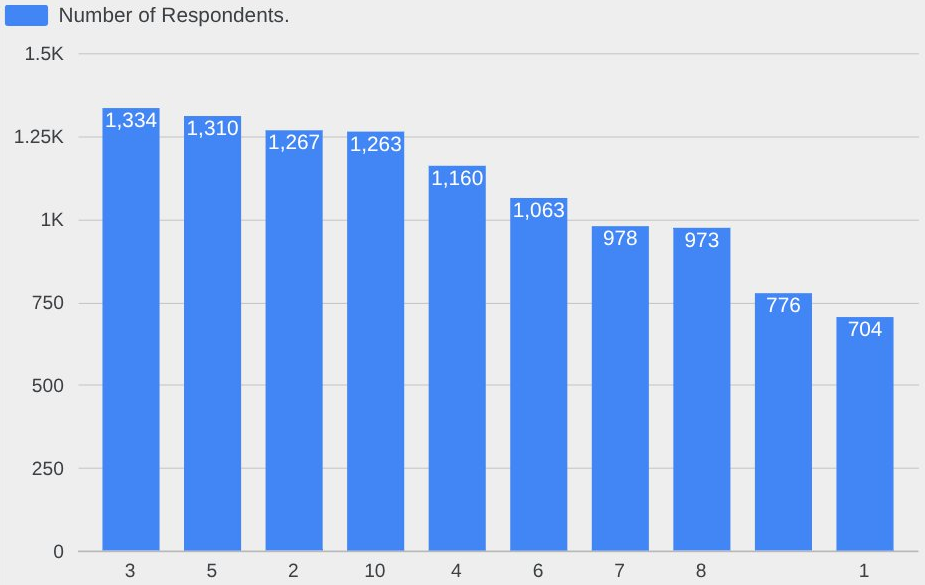
Age Distribution



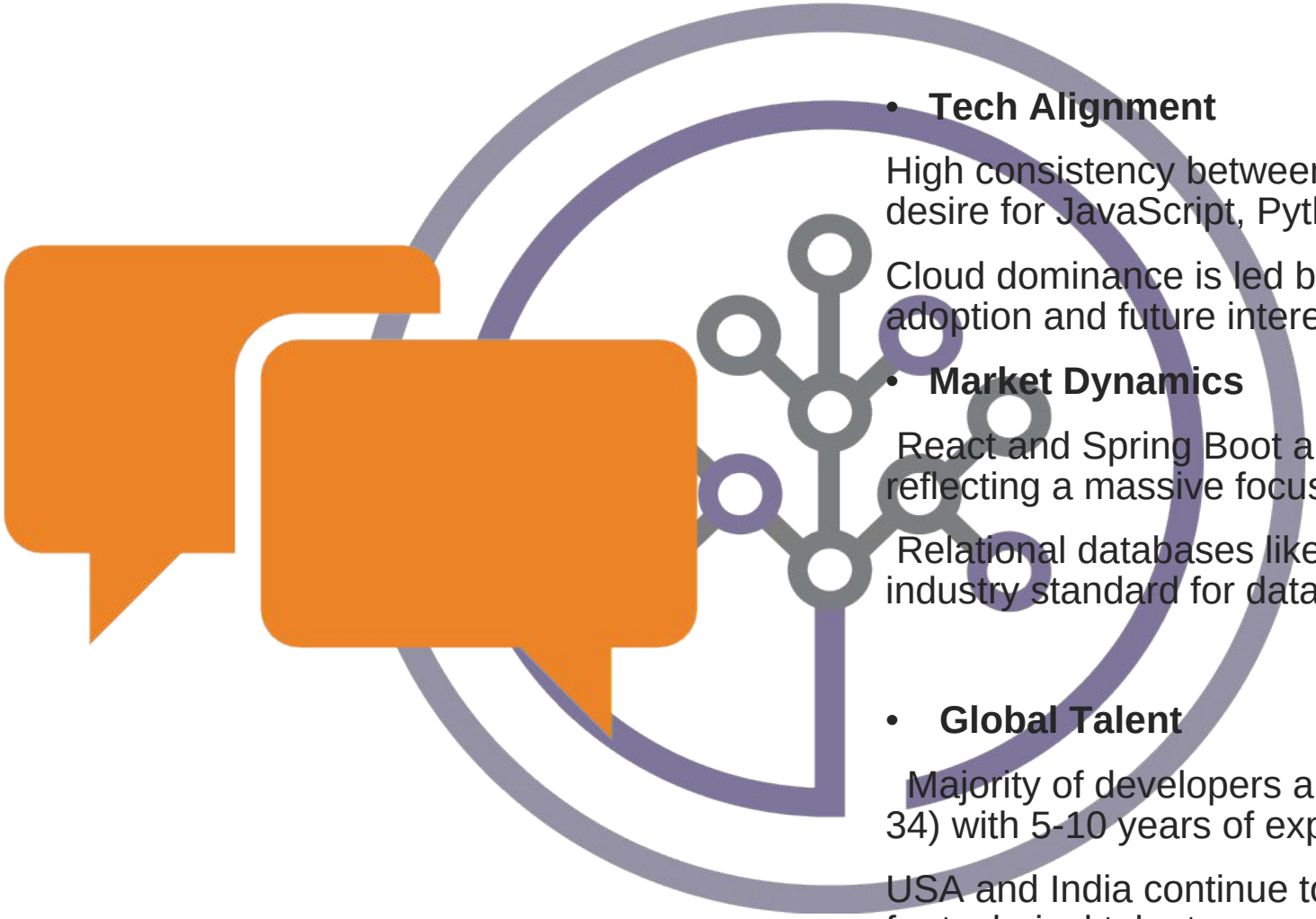
Years of Coding Experience



Years of Professional Coding Experience



DISCUSSION



- **Tech Alignment**

High consistency between current usage and future desire for JavaScript, Python, and PostgreSQL.

Cloud dominance is led by AWS and Azure in both adoption and future interest.

- **Market Dynamics**

React and Spring Boot are the top frameworks, reflecting a massive focus on scalable web apps.

Relational databases like PostgreSQL remain the industry standard for data management.

- **Global Talent**

Majority of developers are young professionals (25-34) with 5-10 years of experience.

USA and India continue to be the primary global hubs for technical talent.

OVERALL FINDINGS & IMPLICATIONS

Findings

- Technology Stability: Python, JavaScript, and PostgreSQL remain the core pillars of the developer ecosystem
- Cloud Centricity: A definitive global shift toward cloud platforms, specifically AWS and Azure, is driving industry growth
- Skilled Workforce: The industry is powered by a highly educated, young professional demographic with significant coding experience

Implications

- Strategic Hiring: Organizations should focus recruitment on developers proficient in "Cloud+Data" ecosystems to stay competitive
- Curriculum Alignment: Educational programs must prioritize scalable frameworks like React and Spring Boot to match market demand
- Global Talent Strategy: Companies can leverage major talent hubs in the USA and India for large-scale technical projects

CONCLUSION

- **Core Takeaways**

Talent Maturity: The global developer pool is experienced and young, ensuring a long-term supply of technical innovation.

Market Stability: The dominance of JavaScript, Python, and SQL remains unchallenged, providing a stable foundation for the software industry

- **Final Recommendations**

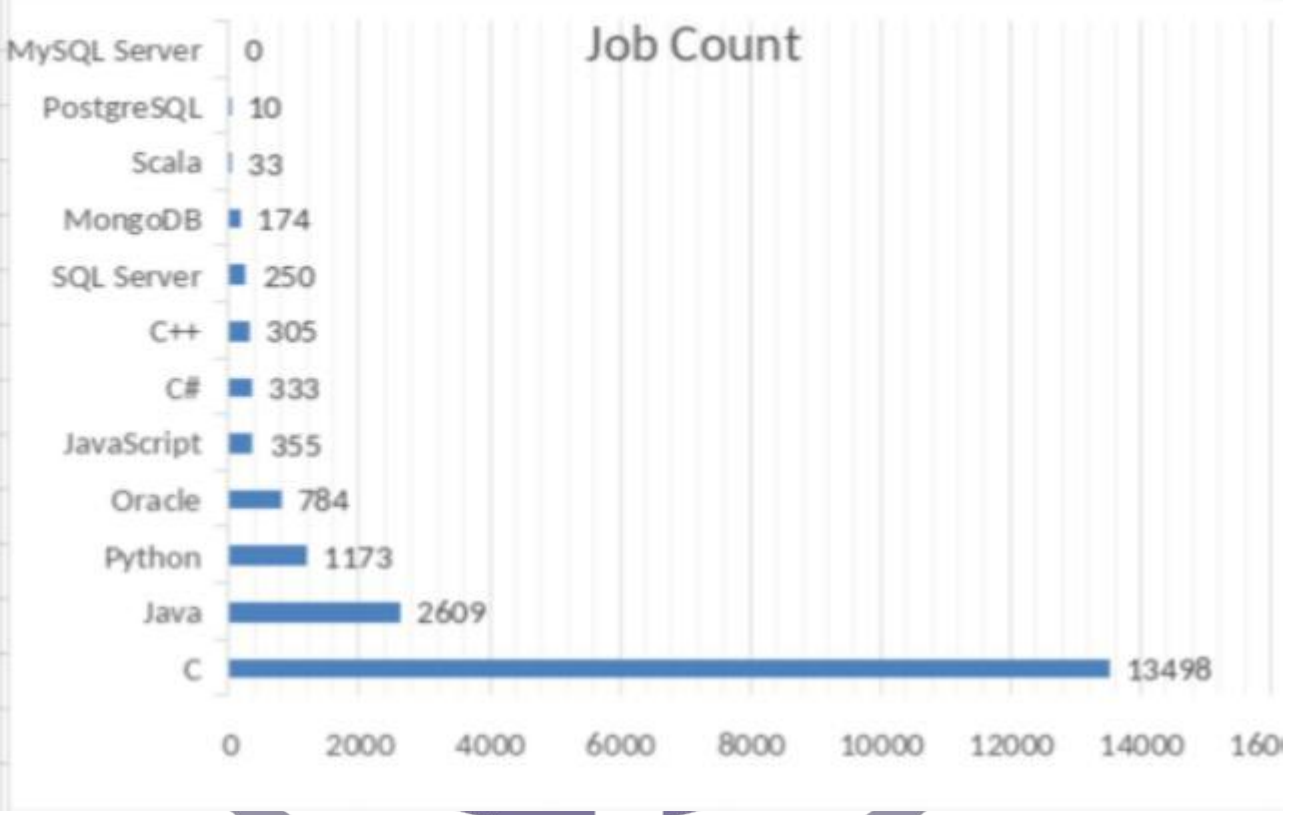
Skill Focus: Individuals should master PostgreSQL and React to align with the highest professional desires.

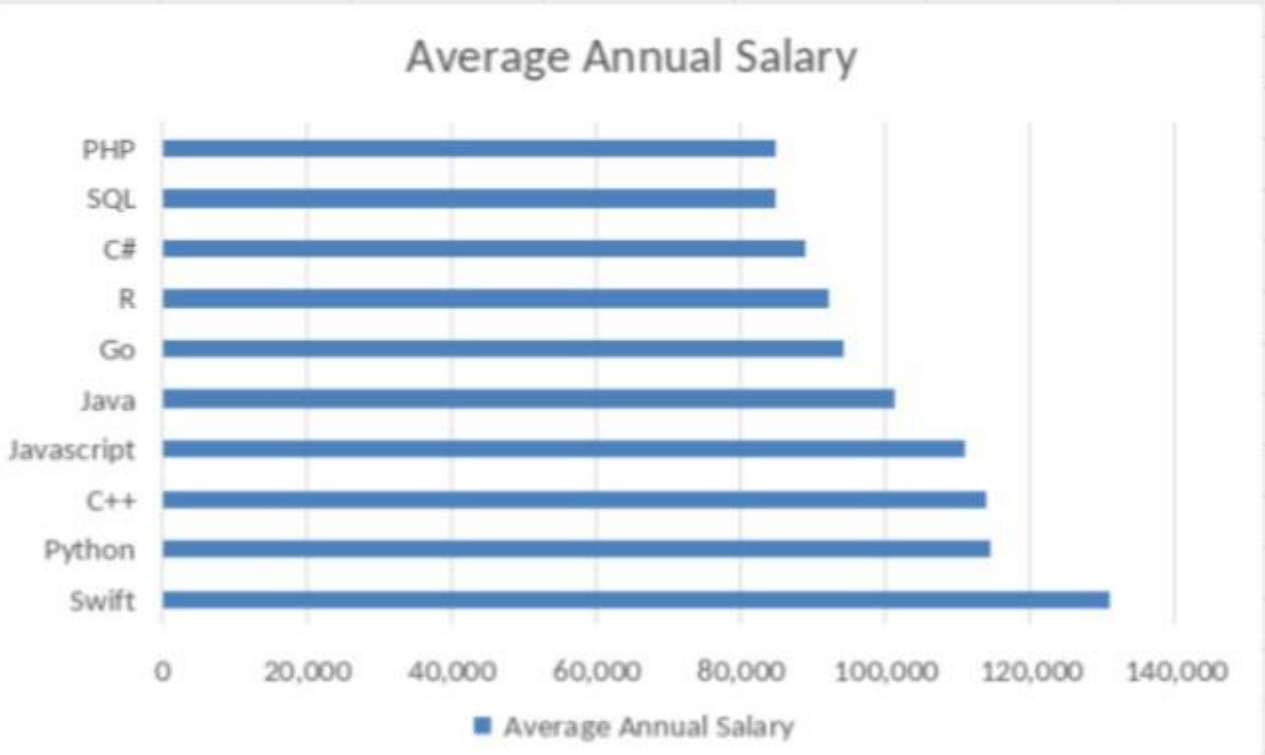
Data-Driven Decisions: Continuous monitoring of developer surveys is essential to adapt to the rapid shifts in technology preferences.

Strategic Investment: Organizations should invest in Cloud-native architectures to leverage the growing trend toward scalable platforms

JOB POSTINGS

	A	B
1	Language	Job Count
2	C	13498
3	Java	2609
4	Python	1173
5	Oracle	784
6	JavaScript	355
7	C#	333
8	C++	305
9	SQL Server	250
10	MongoDB	174
11	Scala	33
12	PostgreSQL	10
13	MySQL Server	0
14		
15		



POPULAR LANGUAGES

Language	Average Annual Salary
Swift	130,801
Python	114,383
C++	113,865
Javascript	110,981
Java	101,013
Go	94,082
R	92,037
C#	88,726
SQL	84,793
PHP	84,727